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EEC 314 — Past Questions & Materials

by Renance

Course: EEC 314

Institution: Delta State Polytechnic, Otefe-Oghara

Papers in this pack: 1

Levels: ND1

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Paper 1 of 1 — ANALOGUE ELECTRONICS III EEC 314 HND1 (20212022) FIRST SEMESTER

Level ND1 • First Semester • Handout

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DELTA STATE POLYTECHNIC, OTEFE-OGHARA

SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL/ELECTRONIC ENGINEERING

FIRST SEMESTER EXAMINATION 2021/2022 Academic Session.

COURSE: ANALOGUE ELECTRONICS III

COURSE CODE: EEC 314

CLASS: HND I

INSTRUCTION: ANSWER ANY FIVE QUESTIONS

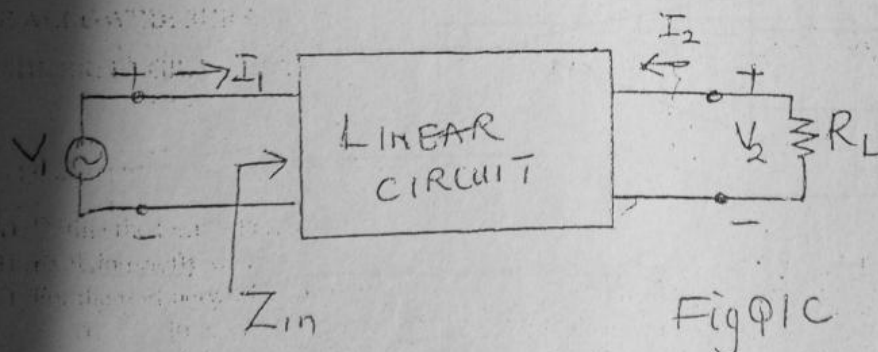
(ALL QUESTIONS CARRY EQUAL MARKS)

TIME ALLOWED: 3HRS

EXAMINER: ENGR. K.O TANNO

QUESTION ONE

- A) Define the term h Parameter (2mks)
- B) Explain briefly why these parameters are called hybrid, or mixed parameter (4mks)
- C) For the port network shown in Fig Q1C below. Derive an expression for
- (i) Input impedance Z_{in} (4mks)
 - (ii) Current gain A_i (2mks)

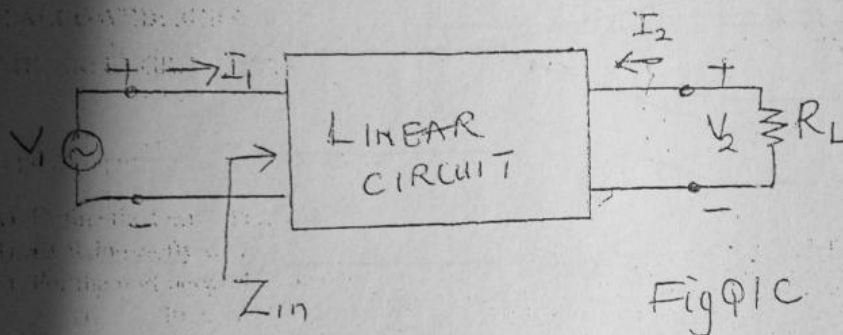


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QUESTION TWO

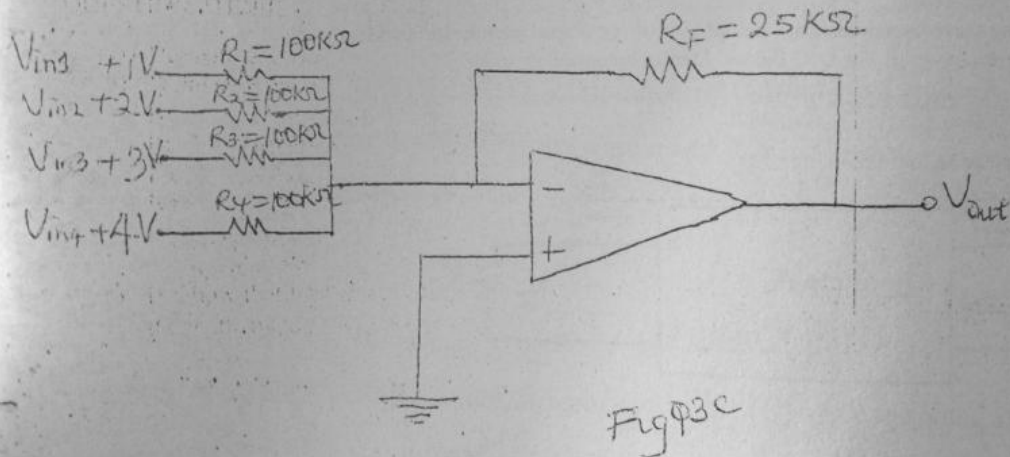
- a) Write short notes on the following terms stating their symbols
- Silicon controlled rectifier (SCR) ($1\frac{1}{2}$ mks)
 - Triac ($1\frac{1}{2}$ mks)
 - Diac ($1\frac{1}{2}$ mks)
- b) Explain briefly the constructional details of the SCR using diagrams where necessary. ($3\frac{1}{2}$ mks)

- c) An ac voltage of $V = 200\sqrt{2}\sin 100\pi t$ is applied to an SCR. If the SCR has a forward breakdown voltage of 200V. Determine the period of conduction. (4Mks).

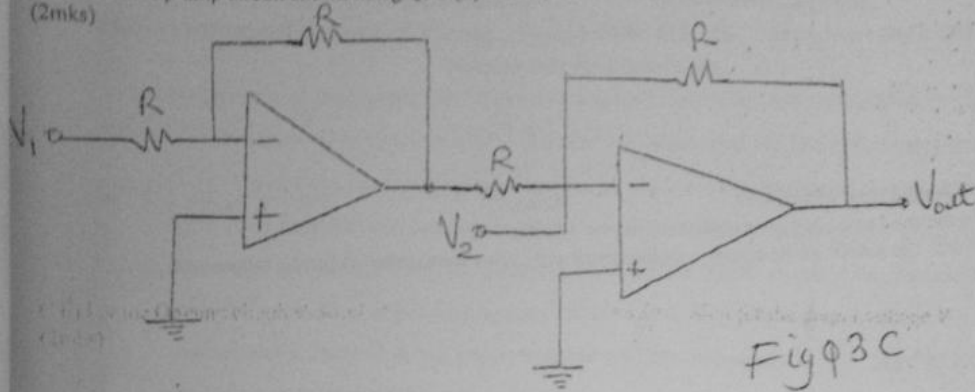
QUESTION THREE.

- a) Define the term common mode rejection ratio (CMRR) as applied to Op amp (2mks)
 b) Write short notes on the following terms
 i) Single ended input (1mk)
 ii) Virtual ground (1mk)
 iii) Differential or double ended input (1mk)
 iv) Common mode input (1mk)

- c) i) Show that for the amplifier in FigQ 3C below produces an output voltage whose magnitude is the mathematical average of the inputs (4mks)

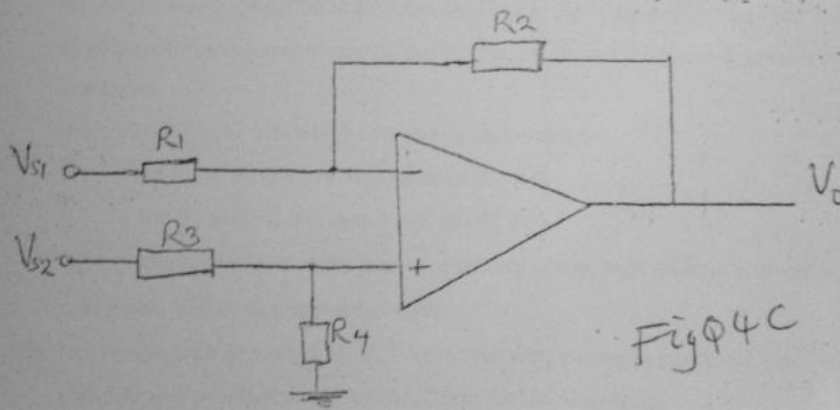


C ii) For the Op-amp circuit shown in fig Q3C (ii) below, write an expression for the output voltage V_o . (2mks)



QUESTION FOUR

- State five assumptions made in the analyzing ideal Op-amp (2½ mks)
- State two disadvantage of the differential amplifier (2mks)
- For the differential amplifier in Fig Q4C, use superposition principle to calculate:-
 - The expression for the output voltage V_o (5mks)
 - If $R_1 = R_3$ and $R_2 = R_4$. Calculate the gain of the differential amplifier. (2½ mks)



QUESTION FIVE

- Define the term logarithmic amplifier (2mks)
- Sketch the base logarithmic amplifier and derive expression for the output voltage. V_o (10mks)

QUESTION SIX

- a) Define the term comparator (2mks)
- b) Sketch the basic antilog amplifier circuit (2mks)
- c) Sketch the circuit of the operational integrator and derive the expression for the output voltage V_o . (8mks)

QUESTION SEVEN

- a) Define the term Noise (2mks)
- b) Sketch the Op-amp differentiator (2mks)
- c) i) Sketch the circuit diagram of the relaxation Oscillator. (6mks)
- ii) If $R = 1k\Omega$, $C = 0.1\mu F$, $R_1 = 18k\Omega$, $R_2 = 2k\Omega$. What is the frequency of the output signal from the oscillator? (2mks)